Day 13 — Genome Annotation

BIOL 4810

2026-03-08

Day 13 — Genome Annotation + Species Identification

Goal: Use **Prokka** (Galaxy) to annotate a bacterial genome assembly, interpret the output files, extract the **16S rRNA** for a quick BLAST-based ID, and then use **TYGS** + **iTOL** to place your genome on a phylogenomic tree.
Submission: Fill in the answers + upload required screenshots/files below → click **Prepare submission preview** → click **Download as PDF (Print)**.

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0) Files you need today

You should already have (from Day 12): - SPAdes assembly contigs file (FASTA)

0A. Assembly file name (as it appears in Galaxy History):

Unk1_SPAdes_contigs.fasta

Local folder (for Git later)

Create a folder inside your BIOL 4810 course folder, for example: - `module2_wgs/day13_prokka_tygs/`

0B. Local folder path/description:

Bioinformatics/Unix_BIOL4810/module2/Annotation

0C) Filter contigs for Prokka (≥500 bp)

Some downstream tools struggle with very short contigs or very large numbers of contigs.

In Galaxy: 1. Search for the tool **Filter FASTA** (it may appear as *Filter sequences by length*). 2. Input: your original SPAdes **contigs FASTA**. 3. Set **minimum sequence length = 500 bp**. 4. Run the tool to create a **filtered contigs** dataset.

Use the filtered contigs for Prokka today.

Keep the original (unfiltered) contigs for TYGS. TYGS can take a long time and you already submitted the full assembly there.

0C. Filtered contigs dataset name (Galaxy History):

Filtered_contigs.fasta

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1) Genome QC (CheckM2 in Galaxy)

Today you will start by evaluating your assembly quality in **Galaxy** using **CheckM2**, which estimates **completeness** and **contamination**.

1A — CheckM2

In Galaxy: 1. Run **CheckM2** on the **filtered assembly FASTA** you will use for Prokka. 2. Open the main CheckM2 results table/report.

CheckM2 tool version (from Job Info):

Check M2 Galaxy Version 1.1.0+galaxy0

Completeness (%):

98.96%

Contamination (%):

0.1%

Interpretation (2–3 sentences): Would you trust this assembly for annotation/species ID? Why?

I would trust this assembly because there is a high percentage of genome completeness and a very low percentage of contamination. From the CheckM2 results, it seems as though we have a trustworthy sample to work with.

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2) Run Prokka on Galaxy

In Galaxy: 1. Search tools for **Prokka** 2. Input: your SPAdes **contigs FASTA** 3. Use defaults unless instructed otherwise 4. Run Prokka

Record your Prokka settings

2A. Prokka version (from Job Info):

Galaxy Version 1.14.6+galaxy1

2B. Any settings changed from defaults? If yes, what?

Model options -> "output quality prediction for both models for each genome"

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3) Prokka outputs — what is each file for?

From Prokka outputs in Galaxy, you will use these most today:

- **.txt** summary statistics
- **.gff** master annotation (features + coordinates)
- **.ffn** nucleotide sequences of gene features (includes rRNA genes)
- **.faa** amino acid sequences of predicted proteins

3A. In your own words, what is the difference between `.gff`, `.ffn`, and `.faa`? (2–4 sentences)

The .gff file has all of the genome information that you could possibly look for. The .ffn file contains data regarding nucleotide sequences of coding genes. Of the legit proteins that our organism makes, the .faa file will keep track of the amino acid sequences of all those coding and hypothetical proteins.

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4) Interpret the Prokka summary (.txt)

Open the Prokka **.txt** summary.

4A. Organism line (paste exactly as reported — it may be "Unknown"):

organism: Genus species strain

4B. contigs:

contigs: 79

4C. bases:

bases: 4797060

4D. CDS:

CDS: 4507

3E. rRNA (total rRNA features):

rRNA: 3

3F. tRNA:

tRNA: 62

3G. tmRNA:

tmRNA: 1

3H. Based on genome size and CDS count, does this look like a typical bacterial genome? Why/why not? (2–3 sentences)

For the most part, I would say yes based on our genome size and CDS count. However, if we were to expand our lens to the rRNA, we have 3 operons instead of 1-2. Although I don't know if this is of concern, I wanted to note that we do have that.

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5) rRNA features (16S/23S/5S) — paste evidence from Prokka

The Prokka `.txt` summary often reports only a total count (e.g., **rRNA: 3**) without listing 16S vs 23S vs 5S in that summary.

To see which rRNAs were found: - Open the **.gff** file and search for `rRNA`, `16S`, `23S`, `5S` **OR** - Open the **.ffn** file and search for `ribosomal RNA`

5A. Paste the description line (header or attribute text) for the 16S rRNA feature:

gnl|Prokka|NCJPFIPA_56 prokka gene 51 1588 . - . ID=NCJPFIPA_04563_gene;locus_tag=NCJPFIPA_04563

5B. Paste the description line (header or attribute text) for the 23S rRNA feature:

gnl|Prokka|NCJPFIPA_53 prokka gene 4 1724 . + . ID=NCJPFIPA_04556_gene;locus_tag=NCJPFIPA_04556

5C. Paste the description line (header or attribute text) for the 5S rRNA feature:

gnl|Prokka|NCJPFIPA_8 prokka gene 252 362 . + . ID=NCJPFIPA_02254_gene;locus_tag=NCJPFIPA_02254

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6) Hypothetical proteins — estimate and interpret

Hypothetical proteins typically appear as: - `product=hypothetical protein` in `.gff` - "hypothetical protein" in headers/annotations in `.faa`

6A. How did you estimate or count hypothetical proteins? (1–2 sentences)

Searched "hypothetical protein" in .gff

6B. Estimated number of hypothetical proteins:

142

6C. Estimated % hypothetical proteins (of CDS):

(142/4507) * 100 = 3.15%

6D. What does a high % hypothetical proteins mean (and NOT mean)? (2–3 sentences)

A high percentage of hypothetical proteins mean that there are proteins that exist within that organism, but we as humans and as scientists do not understand the purpose or function of those proteins yet. More research needs to be done in order to find out those details. Having a high percentage of hypothetical proteins does

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7) Extract 16S rRNA from the Prokka .ffn

Open the Prokka **.ffn** file. Search for a header/annotation indicating **16S ribosomal RNA**.

7A. Paste the FASTA header line for the 16S entry you used:

>NCJPFIPA_04563 16S ribosomal RNA

7B. Paste the full 16S rRNA sequence you used for BLAST (sequence only; you can keep line breaks):

TTGAAGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTC
GAACGGTAACGGGAAGCAGCTTGCTGCTTCGCTGACGAGTGGCGGACGGGTGAGTAATGT
CTGGGAAACTGCCTGATGGAGGGGGATAACTACTGGAAACGGTGGCTAATACCGCATAAC
GTCGCAAGACCAAAGAGGGGGACCTTCGGGCCTCTTGCCATCAGATGTGCCCAGATGGGA
TTAGCTTGTTGGTGAGGTAACGGCTCACCAAGGCGACGATCCCTAGCTGGTCTGAGAGGA
TGACCAGCCACACTGGAACTGAGACACGGTCCAGACTCCTACGGGAGGCAGCAGTGGGGA
ATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCCGCGTGTATGAAGAAGGCCTTCG
GGTTGTAAAGTACTTTCAGCGGGGAGGAAGGTGTTGTGGTTAATAACCGCAGCAATTGAC
GTTACCCGCAGAAGAAGCACCGGCTAACTCCGTGCCAGCAGCCGCGGTAATACGGAGGGT
GCAAGCGTTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTCTGTCAAGTCGGAT

7C. In 1 sentence: why do we use 16S for a quick ID?

We use 16S for a quick ID because every prokaryotic organism has a 16S gene, which makes it easy to pull that from a prokaryote in order to figure out its identity.

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8) 16S BLAST (species ID quick check)

BLAST your 16S sequence: 1. Go to NCBI BLAST 2. Choose **BLASTn** 3. Paste your 16S sequence 4. Run BLAST and choose a reasonable hit to interpret

Upload screenshot: your 16S BLAST results (Descriptions table or Alignment view)

Choose File Screenshot…_results.jpeg

An official website of the United States government Here's how you know

NIH National Library of Medicine
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BLAST® » blastn suite » results for RID-V0PZJF9N016

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Job Title	16S rRNA BLAST
RID	V0PZJF9N016 Search expires on 03-12 01:59 am Download All
Program	BLASTN Citation
Database	rRNA_typestrains/16S_ribosomal_RNA See details
Query ID	lcl|Query_1433003
Description	None
Molecule type	dna
Query Length	1538
Other reports	Distance tree of results MSA viewer

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to E value to Query Coverage to

Filter Reset

Descriptions Graphic Summary Alignments Taxonomy

Sequences producing significant alignments Download Select columns Show 100

select all *0 sequences selected* GenBank Graphics Distance tree of results MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Salmonella enterica subsp. enterica serovar Typhimurium strain LT2 16S ribosomal RNA, partial sequence	Salmonella enter...	2784	2784	100%	0.0	99.48%	1544	NR_074910.1
Salmonella enterica subsp. enterica serovar Typhimurium strain ATCC 13311 16S ribosomal RNA, partial sequen...	Salmonella enter...	2778	2778	100%	0.0	99.15%	1539	NR_119108.1
Salmonella bongori strain NCTC 12419 16S ribosomal RNA, complete sequence	Salmonella bongori	2741	2741	100%	0.0	98.83%	1542	NR_074888.1
Citrobacter telavivensis strain 6105 16S ribosomal RNA, complete sequence	Citrobacter telavi...	2691	2691	100%	0.0	98.24%	1542	NR_180890.1
Salmonella enterica subsp. enterica strain LT2 16S ribosomal RNA, partial sequence	Salmonella enter...	2689	2689	95%	0.0	99.66%	1467	NR_104709.1
Enterobacter cloacae strain ATCC 13047 16S ribosomal RNA, complete sequence	Enterobacter clo...	2652	2652	100%	0.0	97.80%	1543	NR_102794.2
Enterobacter quasiroogenkampii strain WCHECL1060 16S ribosomal RNA, partial sequence	Enterobacter qua...	2652	2652	100%	0.0	97.79%	1538	NR_179166.1
Enterobacter quasihormaechei strain WCHEs120003 16S ribosomal RNA, partial sequence	Enterobacter qua...	2647	2647	100%	0.0	97.72%	1538	NR_180451.1
Enterobacter cloacae strain DSM 30054 16S ribosomal RNA, partial sequence	Enterobacter clo...	2643	2643	99%	0.0	97.85%	1529	NR_117679.1
Enterobacter sichuanensis strain WCHECL1597 16S ribosomal RNA, partial sequence	Enterobacter sic...	2636	2636	99%	0.0	97.78%	1528	NR_179946.1
Enterobacter chengduensis strain WCHECI-C4 16S ribosomal RNA, partial sequence	Enterobacter che...	2636	2636	100%	0.0	97.60%	1538	NR_179167.1
Citrobacter sedlakii strain I-75 16S ribosomal RNA, partial sequence	Citrobacter sedlakii	2632	2632	99%	0.0	97.90%	1522	NR_028686.1
Enterobacter chuandaensis strain 090028 16S ribosomal RNA, partial sequence	Enterobacter chu...	2619	2619	99%	0.0	97.58%	1528	NR_180237.1
Enterobacter cloacae strain 279-56 16S ribosomal RNA, partial sequence	Enterobacter clo...	2615	2615	98%	0.0	97.89%	1511	NR_028912.1
Leclercia adecarboxylata strain CIP 82 92 16S ribosomal RNA, partial sequence	Leclercia adecar...	2612	2612	99%	0.0	97.52%	1527	NR_104933.1
Enterobacter mori strain YIM Hb-3 16S ribosomal RNA, partial sequence	Enterobacter mori	2610	2610	99%	0.0	97.63%	1522	NR_146667.2
Citrobacter farmeri strain CDC 2991-81 16S ribosomal RNA, partial sequence	Citrobacter farmeri	2610	2610	98%	0.0	97.88%	1511	NR_024861.1
Escherichia fergusonii ATCC 35469 16S ribosomal RNA, complete sequence	Escherichia ferg...	2603	2603	100%	0.0	97.21%	1542	NR_074902.1
Enterobacter wuhouensis strain WCHEs120002 16S ribosomal RNA, partial sequence	Enterobacter wu...	2601	2601	100%	0.0	97.21%	1536	NR_180450.1
Enterobacter cloacae subsp. dissolvens strain ATCC 23373 16S ribosomal RNA, partial sequence	Enterobacter clo...	2601	2601	98%	0.0	97.88%	1507	NR_118011.1
Enterobacter cloacae subsp. dissolvens strain LMG 2683 16S ribosomal RNA, partial sequence	Enterobacter clo...	2601	2601	97%	0.0	98.00%	1495	NR_044978.1
Pseudocitrobacter vendiensis strain CPO20170097 16S ribosomal RNA, complete sequence	Pseudocitrobact...	2597	2597	100%	0.0	97.14%	1542	NR_180316.1
Citrobacter amalonaticus strain CECT 863 16S ribosomal RNA, partial sequence	Citrobacter amal...	2590	2590	98%	0.0	97.74%	1504	NR_104823.1
Citrobacter koseri strain LMG 5519 16S ribosomal RNA, partial sequence	Citrobacter koseri	2586	2586	97%	0.0	97.79%	1494	NR_118105.1
Salmonella enterica subsp. salamae strain DSM 9220 16S ribosomal RNA, partial sequence	Salmonella enter...	2586	2586	94%	0.0	98.90%	1448	NR_044372.1
Kosakonia quasisacchari strain WCHEs120001 16S ribosomal RNA, partial sequence	Kosakonia quasi...	2579	2579	100%	0.0	96.95%	1536	NR_169476.1
Pseudenterobacter timonensis strain mt20 16S ribosomal RNA, partial sequence	Pseudenterobact...	2579	2579	98%	0.0	97.54%	1503	NR_179439.1
Citrobacter amalonaticus strain LMG 7873 16S ribosomal RNA, partial sequence	Citrobacter amal...	2579	2579	97%	0.0	97.79%	1494	NR_118106.1
Atlantibacter hermannii strain CIP 103176 16S ribosomal RNA, partial sequence	Atlantibacter her...	2575	2575	96%	0.0	98.11%	1478	NR_104940.1
Salmonella enterica subsp. arizonae strain ATCC 13314 16S ribosomal RNA, partial sequence	Salmonella enter...	2575	2575	97%	0.0	97.92%	1491	NR_041696.1
Klebsiella pneumoniae strain DSM 30104 16S ribosomal RNA, partial sequence	Klebsiella pneum...	2575	2575	100%	0.0	96.94%	1534	NR_036794.1
Enterobacter ludwigii strain EN-119 16S ribosomal RNA, partial sequence	Enterobacter lud...	2573	2573	98%	0.0	97.36%	1513	NR_042349.1
Klebsiella pasteurii strain SPARK836C1 16S ribosomal RNA, complete sequence	Klebsiella pasteurii	2567	2567	100%	0.0	96.82%	1540	NR_180640.1
Citrobacter rodentium strain DO 14784 16S ribosomal RNA, partial sequence	Citrobacter rode...	2567	2567	97%	0.0	97.60%	1496	NR_028685.1
Klebsiella pneumoniae strain DSM 30104 16S ribosomal RNA, partial sequence	Klebsiella pneum...	2566	2566	100%	0.0	96.93%	1530	NR_117686.1
Scandinavium goeteborgense strain CCUG 66741 16S ribosomal RNA, complete sequence	Scandinavium go...	2564	2564	100%	0.0	96.75%	1542	NR_180446.1
Citrobacter werkmanii strain CDC 0876-58 16S ribosomal RNA, partial sequence	Citrobacter werk...	2562	2562	99%	0.0	96.98%	1521	NR_024862.1
Klebsiella pneumoniae strain DSM 30104 16S ribosomal RNA, partial sequence	Klebsiella pneum...	2560	2560	100%	0.0	96.87%	1530	NR_117684.1
Shigella sonnei strain CECT 4887 16S ribosomal RNA, partial sequence	Shigella sonnei	2560	2560	99%	0.0	97.05%	1530	NR_104826.1
Leclercia adecarboxylata ATCC 23216 = NBRC 102595 strain LMG 2803 16S ribosomal RNA, partial sequence	Leclercia adecar...	2560	2560	98%	0.0	97.29%	1506	NR_117405.1
Shigella boydii strain P288 16S ribosomal RNA, partial sequence	Shigella boydii	2558	2558	98%	0.0	97.16%	1515	NR_104901.1
Citrobacter murliniae strain CDC 2970-59 16S ribosomal RNA, partial sequence	Citrobacter murli...	2558	2558	99%	0.0	96.86%	1530	NR_028688.1
Yokenella regensburgei strain CIP 105435 16S ribosomal RNA, partial sequence	Yokenella regens...	2556	2556	99%	0.0	96.86%	1527	NR_104934.1
Superficieibacter electus strain BP-1 16S ribosomal RNA, partial sequence	Superficieibacter...	2555	2555	98%	0.0	97.16%	1512	NR_179954.1
Klebsiella pneumoniae strain DSM 30104 16S ribosomal RNA, partial sequence	Klebsiella pneum...	2555	2555	100%	0.0	96.80%	1530	NR_117685.1
Klebsiella pneumoniae strain DSM 30104 16S ribosomal RNA, partial sequence	Klebsiella pneum...	2555	2555	100%	0.0	96.80%	1530	NR_117683.1
Citrobacter youngae strain GTC 1314 16S ribosomal RNA, partial sequence	Citrobacter youn...	2555	2555	97%	0.0	97.65%	1490	NR_041527.1
Buttiauxella massiliensis strain Marseille-P9829 16S ribosomal RNA, partial sequence	Buttiauxella mas...	2553	2553	100%	0.0	96.63%	1538	NR_181726.1
Cronobacter turicensis z3032 16S ribosomal RNA, complete sequence	Cronobacter turic...	2553	2553	100%	0.0	96.62%	1543	NR_102802.1
Shimwellia blattae DSM 4481 = NBRC 105725 16S ribosomal RNA, partial sequence	Shimwellia blatta...	2553	2553	100%	0.0	96.62%	1544	NR_074908.1
Lelliottia aquatilis strain 6331-17 16S ribosomal RNA, partial sequence	Lelliottia aquatilis	2551	2551	100%	0.0	96.62%	1536	NR_179968.1
Phytobacter palmae strain S29 16S ribosomal RNA, partial sequence	Phytobacter pal...	2547	2547	97%	0.0	97.58%	1487	NR_179034.1
Citrobacter koseri strain CDC-8132-86 16S ribosomal RNA, partial sequence	Citrobacter koseri	2547	2547	96%	0.0	97.89%	1469	NR_104890.1
Salmonella enterica subsp. houtenae strain DSM 9221 16S ribosomal RNA, partial sequence	Salmonella enter...	2545	2545	93%	0.0	98.61%	1437	NR_044371.1
Citrobacter braakii strain 167 16S ribosomal RNA, partial sequence	Citrobacter braakii	2545	2545	99%	0.0	96.73%	1530	NR_028687.1
Klebsiella pneumoniae strain DSM 30104 16S ribosomal RNA, partial sequence	Klebsiella pneum...	2543	2543	99%	0.0	96.79%	1524	NR_117682.1

Feedback

Huaxiibacter chinensis strain 155047 16S ribosomal RNA, partial sequence	Huaxiibacter chin...	2540	2540	100%	0.0	96.49%	1536	NR_184601.1
Cedecea lapagei strain DSM 4587 16S ribosomal RNA, partial sequence	Cedecea lapagei	2540	2540	100%	0.0	96.61%	1532	NR_126317.1
Enterobacter cloacae strain NBRC 13535 16S ribosomal RNA, partial sequence	Enterobacter clo...	2538	2538	95%	0.0	97.89%	1465	NR_113615.1
Kosakonia radicincitans DSM 16656 16S ribosomal RNA, partial sequence	Kosakonia radici...	2532	2532	100%	0.0	96.54%	1530	NR_117704.1
Phytobacter massiliensis JC163 16S ribosomal RNA, partial sequence	Phytobacter mas...	2531	2531	97%	0.0	97.13%	1499	NR_125600.1
Klebsiella aerogenes KCTC 2190 16S ribosomal RNA, complete sequence	Klebsiella aeroge...	2529	2529	100%	0.0	96.36%	1540	NR_102493.2
Erwinia tasmaniensis Et1/99 16S ribosomal RNA, partial sequence	Erwinia tasmanie...	2529	2529	100%	0.0	96.37%	1539	NR_074869.1
Kosakonia sacchari strain SP1 16S ribosomal RNA, partial sequence	Kosakonia sacchari	2529	2529	98%	0.0	97.08%	1500	NR_118333.1
Citrobacter koseri strain CIP 82.87 16S ribosomal RNA, partial sequence	Citrobacter koseri	2523	2523	95%	0.0	97.81%	1459	NR_118588.1
Enterobacter soli ATCC BAA-2102 strain LF7 16S ribosomal RNA, partial sequence	Enterobacter soli...	2523	2523	100%	0.0	96.23%	1535	NR_117547.1
Franconibacter daqui strain DL503 16S ribosomal RNA, partial sequence	Franconibacter d...	2521	2521	98%	0.0	96.95%	1503	NR_159235.1
Cedecea lapagei strain DSM 4587 16S ribosomal RNA, partial sequence	Cedecea lapagei	2521	2521	99%	0.0	96.41%	1531	NR_126319.1
Pantoea phytostimulans strain C3 16S ribosomal RNA, complete sequence	Pantoea phytosti...	2519	2519	100%	0.0	96.24%	1542	NR_199926.1
Cronobacter sakazakii strain ATCC 29544 16S ribosomal RNA, partial sequence	Cronobacter sak...	2519	2519	97%	0.0	97.13%	1493	NR_044076.1
Cedecea lapagei strain DSM 4587 16S ribosomal RNA, partial sequence	Cedecea lapagei	2518	2518	100%	0.0	96.35%	1531	NR_126318.1
Cronobacter muytjensii strain 05CHPL67 16S ribosomal RNA, partial sequence	Cronobacter muy...	2518	2518	98%	0.0	96.82%	1507	NR_117437.1
Escherichia albertii strain Albert 19982 16S ribosomal RNA, partial sequence	Escherichia albertii	2516	2516	97%	0.0	97.06%	1494	NR_025569.1
Escherichia marmotae strain HT073016 16S ribosomal RNA, partial sequence	Escherichia mar...	2512	2512	98%	0.0	96.81%	1504	NR_136472.1
Tatumella punctata strain SHS 2003 16S ribosomal RNA, partial sequence	Tatumella punctata	2510	2510	99%	0.0	96.34%	1527	NR_104937.1
Citrobacter freundii ATCC 8090 = MTCC 1658 = NBRC 12681 16S ribosomal RNA, partial sequence	Citrobacter freun...	2510	2510	98%	0.0	96.87%	1505	NR_028894.1
Citrobacter gillenii strain CDC 4693-86 16S ribosomal RNA, partial sequence	Citrobacter gillenii	2510	2510	99%	0.0	96.40%	1527	NR_041697.1
Shigella flexneri strain ATCC 29903 16S ribosomal RNA, partial sequence	Shigella flexneri	2510	2510	97%	0.0	97.11%	1488	NR_026331.1
Siccibacter turicensis LMG 23730 16S ribosomal RNA, partial sequence	Siccibacter turice...	2507	2507	97%	0.0	96.92%	1494	NR_104924.1
Klebsiella planticola ATCC 33531 strain JCM 7251 16S ribosomal RNA, partial sequence	Klebsiella planti...	2507	2507	94%	0.0	97.80%	1451	NR_112011.1
Pantoea stewartii subsp. indologenes strain CIP 104006 16S ribosomal RNA, partial sequence	Pantoea stewartii...	2505	2505	99%	0.0	96.33%	1524	NR_104928.1
Citrobacter pasteurii strain CIP 55.13 16S ribosomal RNA, partial sequence	Citrobacter paste...	2503	2503	97%	0.0	96.92%	1492	NR_178769.1
Shimwellia blattae DSM 4481 = NBRC 105725 strain CIP 104942 16S ribosomal RNA, partial sequence	Shimwellia blatta...	2503	2503	99%	0.0	96.33%	1525	NR_118256.1
Salmonella enterica subsp. arizonae strain DSM 9386 16S ribosomal RNA, partial sequence	Salmonella enter...	2501	2501	93%	0.0	98.05%	1438	NR_116125.1
Erwinia psidii strain LMG 7039 16S ribosomal RNA, partial sequence	Erwinia psidii	2499	2499	97%	0.0	96.85%	1494	NR_104569.1
Pectobacterium polaris strain NIBIO1006 16S ribosomal RNA, complete sequence	Pectobacterium...	2497	2497	100%	0.0	95.98%	1542	NR_159084.1
Mixta intestinalis strain 29Y89B 16S ribosomal RNA, partial sequence	Mixta intestinalis	2497	2497	97%	0.0	96.79%	1502	NR_137380.1
Klebsiella pneumoniae strain DSM 30104 16S ribosomal RNA, partial sequence	Klebsiella pneum...	2497	2497	97%	0.0	96.79%	1492	NR_114715.1
Leclercia adecarboxylata strain NBRC 102595 16S ribosomal RNA, partial sequence	Leclercia adecar...	2494	2494	95%	0.0	97.35%	1465	NR_114154.1
Klebsiella oxytoca strain ATCC 13182 16S ribosomal RNA, partial sequence	Klebsiella oxytoca	2494	2494	98%	0.0	96.61%	1502	NR_118853.1
Pectobacterium polaris strain NIBIO1006 16S ribosomal RNA, complete sequence	Pectobacterium...	2492	2492	100%	0.0	95.91%	1542	NR_159083.1
Klebsiella quasivariicola strain KPN1705 16S ribosomal RNA, partial sequence	Klebsiella quasiv...	2492	2492	98%	0.0	96.55%	1505	NR_181901.1
Erwinia mallotivora strain DSM 4565 16S ribosomal RNA, partial sequence	Erwinia mallotivora	2490	2490	98%	0.0	96.55%	1504	NR_041974.1
Pectobacterium polaris strain NIBIO1392 16S ribosomal RNA, partial sequence	Pectobacterium...	2488	2488	100%	0.0	95.96%	1534	NR_159086.1
Mixta mediterraneensis strain Marseille-Q2057 16S ribosomal RNA, partial sequence	Mixta mediterran...	2486	2486	100%	0.0	95.85%	1538	NR_181487.1
Salmonella enterica subsp. indica strain DSM 14848 16S ribosomal RNA, partial sequence	Salmonella enter...	2486	2486	92%	0.0	98.38%	1414	NR_044370.1
Enterobacter cancerogenus strain LMG 2693 16S ribosomal RNA, partial sequence	Enterobacter can...	2486	2486	97%	0.0	96.53%	1495	NR_044977.1
Pectobacterium parmentieri strain RNS 08-42-1A 16S ribosomal RNA, complete sequence	Pectobacterium...	2481	2481	100%	0.0	95.78%	1543	NR_153752.1
Pectobacterium quasiaquaticum strain A477-S1-J17 16S ribosomal RNA, complete sequence	Pectobacterium...	2481	2481	100%	0.0	95.78%	1542	NR_181464.1
Buttiauxella agrestis ATCC 33320 strain DSM 4586 16S ribosomal RNA, partial sequence	Buttiauxella agre...	2481	2481	98%	0.0	96.43%	1504	NR_041968.1

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8A. Top hit organism name (as shown):

Salmonella enterica subspp. enterica serovar Typhimurium strain

8B. % identity and query coverage for that hit:

99.48% identity, 100% query coverage

8C. Your best interpretation of species/genus (be cautious) (2–3 sentences):

My best interpretation of the species is that it is some kind of Salmonella, even though we had quite a few Enterobacter species pop up right after the Salmonella species. The top two Salmonella species were derivatives of Salmonella enterica, just with different subspecies. Data such as the query cover, percent

8D. One reason 16S BLAST can be misleading or insufficient (1–2 sentences):

16S BLAST can be misleading because we are only looking at a single gene rather than a whole or complete genome. We could get more information with a whole genome.

---

9) ANI confirmation (EZBioCloud ANI)

After you have a likely species ID from **16S BLAST**, use ANI as a stronger genome-wide comparison.

You will use the **EZBioCloud ANI** tool here (open in a new tab): https://www.ezbiocloud.net/tools/ani (https://www.ezbiocloud.net/tools/ani)

What you will do 1. Download your **FILTERED contigs FASTA** (the same one you used for Prokka today). 2. Download **two reference genomes** (FASTA) from NCBI for *two candidate species* you want to compare against (often the top 16S BLAST hit and one close alternative). 3. Run ANI twice (your genome vs reference genome 1; your genome vs reference genome 2).

9A. Species/strain for comparison #1 (your chosen reference):

Salmonella enterica subsp. enterica serovar Typhimurium

9B. ANI for comparison #1 (%):

99.1%

9C. Species/strain for comparison #2 (your chosen reference):

Salmonella bongori

9D. ANI for comparison #2 (%):

98.7%

9E. Interpretation (2–3 sentences): Which comparison supports the strongest species-level match, and why?

The first comparison with Salmonella enterica subsp. enterica serovar Typhimurium is the strongest species-level match because the average nucleotide identity was almost perfect. Because the ANI is greater than 95%, we can infer that this comparison has the most likely chance of being the species of the unknown bacterium we

---

10) TYGS — what it provides

Submit your assembly to TYGS (Type Strain Genome Server). TYGS will: - Compare your genome to type strains - Provide **dDDH** (species boundary ~70%) - Produce a phylogenomic tree

10A. In your own words: what is TYGS doing that 16S BLAST is not? (2–4 sentences)

TYGS is analyzing the entire genomes of organisms, whereas 16S BLAST is only looking at a single gene. TYGS also gives us the oppurtunity to build a phylogenetic tree based on the protein or DNA sequence.

10B. Record your closest type strain match (name) and dDDH value (if available today):

Salmonella typhimurium JCM 1652

10C. What does the 70% dDDH threshold mean? (1–2 sentences)

The 70% dDDH threshold means that if the dDDH value is equal to a greater than 70%, the identity of the species is the same as what is predicted. If the dDDH value is under 70%, then the unknown bacterium is likely

10D. If TYGS shows low branch support and offers a proteome-based tree, what does that imply? (2–3 sentences)

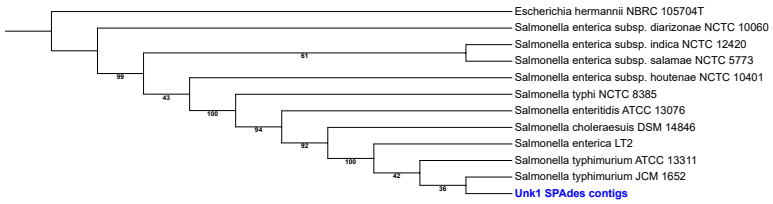
TYGS showing low branch support and offering a proteome-based tree implies that the evolutionary connections between the species on the tree are very close or there is not enough data to sufficiently draw a good proteome-based tree. The genomic data between species is either super close to where TYGS cannot tell the difference between species or the proteome difference is too large that TYGS tries to artificially fill in the

11) iTOL — visualize your TYGS tree

1. From TYGS, download the **tree file** (**.phy**)
2. Go to **iTOL** (Interactive Tree of Life)
3. Upload the tree
4. Adjust labels/formatting enough to be readable
5. Take a screenshot

Upload screenshot: iTOL tree visualization (from TYGS tree)

Choose File Salmonella_…_SVG_.svg



11A. What does the tree tell you that a single BLAST hit does not? (2–3 sentences)

The tree is able to tell me the bootstrap values of multiple species whereas the single BLAST hit does not do that. From the bootstrap values, we can make predictions on how likely an evolutionary pathway was formed.

11B. Identify one branch you trust and one you would be cautious about. Explain using support values if shown. (2–4 sentences)

I am not too confident in the branch right next to the Unk1_SPAdes_contigs, which belongs to Salmonella typhimurium JCM 1652 because the bootstrap values for their relationship to their common ancestor are below 40%. That tells me that my unknown species and S. thyphimurium might not be that closely related as I thought. On the other hand, I did see a branch that had a bootstrap value of 100% between Salmonella

---

12) What to download and save for Git

Minimum recommended downloads for your local folder: - Prokka **.txt** summary - Prokka **.gff** - Prokka **.ffn** - Prokka **.faa** - CheckM2 results/report (table) - TYGS tree file (**.phy**) - Your PDF from this page (print-to-PDF)

12A. Write 4–6 lines of a "workflow note" you would put in `workflow_day13.md` (include tool versions when possible).

Parameters: Whole-Proteome tree

6. Visualize phylogenetic tree; Tool: interactive Tree of Life (iTOL) - iTOL v7 (web), display bootstraps as text, rotated children, bold "Unk1_SPAdes contigs"

7. Avg. Nucleotide ID (ANI); Tool: EZ BioCloud (Web), from NCBI genome database, downloaded the FASTA

Export (PDF)

1. Upload the required screenshots above (BLAST + iTOL)
2. Click **Prepare submission preview**
3. Click **Download as PDF (Print)**

Submission preview

Setup
Assembly file: Unk1_SPAdes_contigs.fasta
Local folder: Bioinformatics/Unix_BIOL4810/module2/Annotation

Prokka run
Version: Galaxy Version 1.14.6+galaxy1
Settings: Model options -> "output quality prediction for both models for each genome"

Outputs understanding
.gff vs .ffn vs .faa: The .gff file has all of the genome information that you could possibly look for. The .ffn file contains data regarding nucleotide sequences of coding genes. Of the legit proteins that our organism makes, the .faa file will keep track of the amino acid sequences of all those coding and hypothetical proteins.

Prokka summary (.txt)
Organism: organism: Genus species strain
Contigs: contigs: 79 *Bases:* bases: 4797060
CDS: CDS: 4507 *rRNA:* rRNA: 3 *tRNA:* tRNA: 62 *tmRNA:* tmRNA: 1
Sanity check: For the most part, I would say yes based on our genome size and CDS count. However, if we were to expand our lens to the rRNA, we have 3 operons instead of 1-2. Although I don't know if this is of concern, I wanted to note that we do have that.

rRNA evidence
16S description: gnl|Prokka|NCJPFIPA_56 prokka gene 51 1588 . - . ID=NCJPFIPA_04563_gene;locus_tag=NCJPFIPA_04563 gnl|Prokka|NCJPFIPA_56 barrnap:0.9 rRNA 51 1588 0 - . ID=NCJPFIPA_04563;Parent=NCJPFIPA_04563_gene;locus_tag=NCJPFIPA_04563;product=16S ribosomal RNA
23S description: gnl|Prokka|NCJPFIPA_53 prokka gene 4 1724 . + . ID=NCJPFIPA_04556_gene;locus_tag=NCJPFIPA_04556 gnl|Prokka|NCJPFIPA_53 barrnap:0.9 rRNA 4 1724 0 + . ID=NCJPFIPA_04556;Parent=NCJPFIPA_04556_gene;locus_tag=NCJPFIPA_04556;note=aligned only 53 percent of the 23S ribosomal RNA;product=23S ribosomal RNA (partial)
5S description: gnl|Prokka|NCJPFIPA_8 prokka gene 252 362 . + . ID=NCJPFIPA_02254_gene;locus_tag=NCJPFIPA_02254 gnl|Prokka|NCJPFIPA_8 barrnap:0.9 rRNA 252 362 7.3e-11 + . ID=NCJPFIPA_02254;Parent=NCJPFIPA_02254_gene;locus_tag=NCJPFIPA_02254;product=5S ribosomal RNA

CheckM2
CheckM2 version: Check M2 Galaxy Version 1.1.0+galaxy0
Completeness: 98.96%% *Contamination:* 0.1%%
CheckM2 interpretation: I would trust this assembly because there is a high percentage of genome completeness and a very low percentage of contamination. From the CheckM2 results, it seems as though we have a trustworthy sample to work with.

Hypothetical proteins
How estimated: Searched "hypothetical protein" in .gff
 # hypothetical: 142 *% hypothetical:* (142/4507) * 100 = 3.15%
Interpretation: A high percentage of hypothetical proteins mean that there are proteins that exist within that organism, but we as humans and as scientists do not understand the purpose or function of those proteins yet. More research needs to be done in order to find out those details. Having a high percentage of hypothetical proteins does NOT mean that the proteins are not worth anything.

16S extraction
16S header: >NCJPFIPA_04563 16S ribosomal RNA
16S sequence:

```
TTGAAGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTC
GAACGGTAACGGGAAGCAGCTTGCTGCTTCGCTGACGAGTGGCGGACGGGTGAGTAATGT
CTGGGAAACTGCCTGATGGAGGGGGATAACTACTGGAAACGGTGGCTAATACCGCATAAC
GTCGCAAGACCAAAGAGGGGGACCTTCGGGCCTCTTGCCATCAGATGTGCCCAGATGGGA
TTAGCTTGTTGGTGAGGTAACGGCTCACCAAGGCGACGATCCCTAGCTGGTCTGAGAGGA
TGACCAGCCACACTGGAACTGAGACACGGTCCAGACTCCTACGGGAGGCAGCAGTGGGGA
ATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCCGCGTGTATGAAGAAGGCCTTCG
GGTTGTAAAGTACTTTCAGCGGGGAGGAAGGTGTTGTGGTTAATAACCGCAGCAATTGAC
GTTACCCGCAGAAGAAGCACCGGCTAACTCCGTGCCAGCAGCCGCGGTAATACGGAGGGT
GCAAGCGTTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTCTGTCAAGTCGGAT
GTGAAATCCCCGGGCTCAACCTGGGAACTGCATTCGAAACTGGCAGGCTTGAGTCTTGTA
GAGGGGGGTAGAATTCCAGGTGTAGCGGTGAAATGCGTAGAGATCTGGAGGAATACCGGT
GGCGAAGGCGGCCCCCTGGACAAAGACTGACGCTCAGGTGCGAAAGCGTGGGGAGCAAAC
AGGATTAGATACCCTGGTAGTCCACGCCGTAAACGATGTCTACTTGGAGGTTGTGCCCTT
GAGGCGTGGCTTCCGGAGCTAACGCGTTAAGTAGACCGCCTGGGGAGTACGGCCGCAAGG
TTAAAACTCAAATGAATTGACGGGGGCCCGCACAAGCGGTGGAGCATGTGGTTTAATTCG
ATGCAACGCGAAGAACCTTACCTGGTCTTGACATCCACAGAAGAATCCAGAGATGGATTT
GTGCCTTCGGGAACTGTGAGACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTTGTGAAAT
GTTGGGTTAAGTCCCGCAACGAGCGCAACCCTTATCCTTTGTTGCCAGCGATTAGGTCGG
GAACTCAAAGGAGACTGCCAGTGATAAACTGGAGGAAGGTGGGGATGACGTCAAGTCATC
ATGGCCCTTACGACCAGGGCTACACACGTGCTACAATGGCGCATACAAAGAGAAGCGACC
TCGCGAGAGCAAGCGGACCTCATAAAGTGCGTCGTAGTCCGGATTGGAGTCTGCAACTCG
ACTCCATGAAGTCGGAATCGCTAGTAATCGTGGATCAGAATGCCACGGTGAATACGTTCC
CGGGCCTTGTACACACCGCCCGTCACACCATGGGAGTGGGTTGCAAAAGAAGTAGGTAGC
TTAACCTTCGGGAGGGCGCTTACCACTTTGTGATTCATGACTGGGGTGAAGTCGTAACAA
GGTAACCGTAGGGGAACCTGCGGTTGGATCACCTCCTT
```

Why 16S: We use 16S for a quick ID because every prokaryotic organism has a 16S gene, which makes it easy to pull that from a prokaryote in order to figure out its identity.

16S BLAST
Top hit: Salmonella enterica subspp. enterica serovar Typhimurium strain
Metrics: 99.48% identity, 100% query coverage
Interpretation: My best interpretation of the species is that it is some kind of Salmonella, even though we had quite a few Enterobacter species pop up right after the Salmonella species. The top two Salmonella species were derivatives of Salmonella enterica, just with different subspecies. Data such as the query cover, percent identity, max score, and the E value suggest that Salmonella enterica is our most likely bacterium.

Limitation: 16S BLAST can be misleading because we are only looking at a single gene rather than a whole or complete genome. We could get more information with a whole genome.

BLAST screenshot

An official website of the United States government Here's how you know

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BLAST ® » blastn suite » results for RID-V0PZJF9N016

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Job Title	16S rRNA BLAST
RID	V0PZJF9N016 Search expires on 03-12 01:59 am Download All
Program	BLASTN Citation
Database	rRNA_typestrains/16S_ribosomal_RNA See details
Query ID	lcl|Query_1433003
Description	None
Molecule type	dna
Query Length	1538
Other reports	Distance tree of results MSA viewer

Filter Results

Organism only top 20 will appear exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to E value to Query Coverage to

Filter Reset

Descriptions Graphic Summary Alignments Taxonomy

Sequences producing significant alignments Download Select columns Show 100

select all 0 sequences selected GenBank Graphics Distance tree of results MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Salmonella enterica subsp. enterica serovar Typhimurium strain LT2 16S ribosomal RNA, partial sequence	Salmonella enter...	2784	2784	100%	0.0	99.48%	1544	NR_074910.1
Salmonella enterica subsp. enterica serovar Typhimurium strain ATCC 13311 16S ribosomal RNA, partial sequen...	Salmonella enter...	2778	2778	100%	0.0	99.15%	1539	NR_119108.1
Salmonella bongori strain NCTC 12419 16S ribosomal RNA, complete sequence	Salmonella bongori	2741	2741	100%	0.0	98.83%	1542	NR_074888.1
Citrobacter telavivensis strain 6105 16S ribosomal RNA, complete sequence	Citrobacter telavi...	2691	2691	100%	0.0	98.24%	1542	NR_180890.1
Salmonella enterica subsp. enterica strain LT2 16S ribosomal RNA, partial sequence	Salmonella enter...	2689	2689	95%	0.0	99.66%	1467	NR_104709.1
Enterobacter cloacae strain ATCC 13047 16S ribosomal RNA, complete sequence	Enterobacter clo...	2652	2652	100%	0.0	97.80%	1543	NR_102794.2
Enterobacter quasiroggenkampii strain WCHECL1060 16S ribosomal RNA, partial sequence	Enterobacter qua...	2652	2652	100%	0.0	97.79%	1538	NR_179166.1
Enterobacter quasihormaechei strain WCHEs120003 16S ribosomal RNA, partial sequence	Enterobacter qua...	2647	2647	100%	0.0	97.72%	1538	NR_180451.1
Enterobacter cloacae strain DSM 30054 16S ribosomal RNA, partial sequence	Enterobacter clo...	2643	2643	99%	0.0	97.85%	1529	NR_117679.1
Enterobacter sichuanensis strain WCHECL1597 16S ribosomal RNA, partial sequence	Enterobacter sic...	2636	2636	99%	0.0	97.78%	1528	NR_179946.1
Enterobacter chengduensis strain WCHECI-C4 16S ribosomal RNA, partial sequence	Enterobacter che...	2636	2636	100%	0.0	97.60%	1538	NR_179167.1
Citrobacter sedlakii strain I-75 16S ribosomal RNA, partial sequence	Citrobacter sedlakii	2632	2632	99%	0.0	97.90%	1522	NR_028686.1
Enterobacter chuandaensis strain 090028 16S ribosomal RNA, partial sequence	Enterobacter chu...	2619	2619	99%	0.0	97.58%	1528	NR_180237.1
Enterobacter cloacae strain 279-56 16S ribosomal RNA, partial sequence	Enterobacter clo...	2615	2615	98%	0.0	97.89%	1511	NR_028912.1
Leclercia adecarboxylata strain CIP 82.92 16S ribosomal RNA, partial sequence	Leclercia adecar...	2612	2612	99%	0.0	97.52%	1527	NR_104933.1
Enterobacter mori strain YIM Hb-3 16S ribosomal RNA, partial sequence	Enterobacter mori	2610	2610	99%	0.0	97.63%	1522	NR_146667.2
Citrobacter farmeri strain CDC 2991-81 16S ribosomal RNA, partial sequence	Citrobacter farmeri	2610	2610	98%	0.0	97.88%	1511	NR_024861.1
Escherichia fergusonii ATCC 35469 16S ribosomal RNA, complete sequence	Escherichia ferg...	2603	2603	100%	0.0	97.21%	1542	NR_074902.1
Enterobacter wuhouensis strain WCHEs120002 16S ribosomal RNA, partial sequence	Enterobacter wu...	2601	2601	100%	0.0	97.21%	1536	NR_180450.1
Enterobacter cloacae subsp. dissolvens strain ATCC 23373 16S ribosomal RNA, partial sequence	Enterobacter clo...	2601	2601	98%	0.0	97.88%	1507	NR_118011.1
Enterobacter cloacae subsp. dissolvens strain LMG 2683 16S ribosomal RNA, partial sequence	Enterobacter clo...	2601	2601	97%	0.0	98.00%	1495	NR_044978.1
Pseudocitrobacter vendiensis strain CPO20170097 16S ribosomal RNA, complete sequence	Pseudocitrobact...	2597	2597	100%	0.0	97.14%	1542	NR_180316.1
Citrobacter amalonaticus strain CECT 863 16S ribosomal RNA, partial sequence	Citrobacter amal...	2590	2590	98%	0.0	97.74%	1504	NR_104823.1
Citrobacter koseri strain LMG 5519 16S ribosomal RNA, partial sequence	Citrobacter koseri	2586	2586	97%	0.0	97.79%	1494	NR_118105.1
Salmonella enterica subsp. salamae strain DSM 9220 16S ribosomal RNA, partial sequence	Salmonella enter...	2586	2586	94%	0.0	98.90%	1448	NR_044372.1
Kosakonia quasisacchari strain WCHEs120001 16S ribosomal RNA, partial sequence	Kosakonia quasi...	2579	2579	100%	0.0	96.95%	1536	NR_169476.1
Pseudenterobacter timonensis strain mt20 16S ribosomal RNA, partial sequence	Pseudenterobact...	2579	2579	98%	0.0	97.54%	1503	NR_179439.1
Citrobacter amalonaticus strain LMG 7873 16S ribosomal RNA, partial sequence	Citrobacter amal...	2579	2579	97%	0.0	97.79%	1494	NR_118106.1
Atlantibacter hermannii strain CIP 103176 16S ribosomal RNA, partial sequence	Atlantibacter her...	2575	2575	96%	0.0	98.11%	1478	NR_104940.1
Salmonella enterica subsp. arizonae strain ATCC 13314 16S ribosomal RNA, partial sequence	Salmonella enter...	2575	2575	97%	0.0	97.92%	1491	NR_041696.1
Klebsiella pneumoniae strain DSM 30104 16S ribosomal RNA, partial sequence	Klebsiella pneum...	2575	2575	100%	0.0	96.94%	1534	NR_036794.1
Enterobacter ludwigii strain EN-119 16S ribosomal RNA, partial sequence	Enterobacter lud...	2573	2573	98%	0.0	97.36%	1513	NR_042349.1
Klebsiella pasteurii strain SPARK836C1 16S ribosomal RNA, complete sequence	Klebsiella pasteurii	2567	2567	100%	0.0	96.82%	1540	NR_180640.1
Citrobacter rodentium strain DO 14784 16S ribosomal RNA, partial sequence	Citrobacter rode...	2567	2567	97%	0.0	97.60%	1496	NR_028685.1
Klebsiella pneumoniae strain DSM 30104 16S ribosomal RNA, partial sequence	Klebsiella pneum...	2566	2566	100%	0.0	96.93%	1530	NR_117686.1
Scandinavium goeteborgense strain CCUG 66741 16S ribosomal RNA, complete sequence	Scandinavium go...	2564	2564	100%	0.0	96.75%	1542	NR_180446.1
Citrobacter werkmanii strain CDC 0876-58 16S ribosomal RNA, partial sequence	Citrobacter werk...	2562	2562	99%	0.0	96.98%	1521	NR_024862.1
Klebsiella pneumoniae strain DSM 30104 16S ribosomal RNA, partial sequence	Klebsiella pneum...	2560	2560	100%	0.0	96.87%	1530	NR_117684.1
Shigella sonnei strain CECT 4887 16S ribosomal RNA, partial sequence	Shigella sonnei	2560	2560	99%	0.0	97.05%	1530	NR_104826.1
Leclercia adecarboxylata ATCC 23216 = NBRC 102595 strain LMG 2803 16S ribosomal RNA, partial sequence	Leclercia adecar...	2560	2560	98%	0.0	97.29%	1506	NR_117405.1
Shigella boydii strain P288 16S ribosomal RNA, partial sequence	Shigella boydii	2558	2558	98%	0.0	97.16%	1515	NR_104901.1
Citrobacter murliniae strain CDC 2970-59 16S ribosomal RNA, partial sequence	Citrobacter murli...	2558	2558	99%	0.0	96.86%	1530	NR_028688.1
Yokenella regensburgei strain CIP 105435 16S ribosomal RNA, partial sequence	Yokenella regens...	2556	2556	99%	0.0	96.86%	1527	NR_104934.1
Superficieibacter electus strain BP-1 16S ribosomal RNA, partial sequence	Superficieibacter...	2555	2555	98%	0.0	97.16%	1512	NR_179954.1
Klebsiella pneumoniae strain DSM 30104 16S ribosomal RNA, partial sequence	Klebsiella pneum...	2555	2555	100%	0.0	96.80%	1530	NR_117685.1
Klebsiella pneumoniae strain DSM 30104 16S ribosomal RNA, partial sequence	Klebsiella pneum...	2555	2555	100%	0.0	96.80%	1530	NR_117683.1
Citrobacter youngae strain GTC 1314 16S ribosomal RNA, partial sequence	Citrobacter youn...	2555	2555	97%	0.0	97.65%	1490	NR_041527.1
Buttiauxella massiliensis strain Marseille-P9829 16S ribosomal RNA, partial sequence	Buttiauxella mas...	2553	2553	100%	0.0	96.63%	1538	NR_181726.1
Cronobacter turicensis z3032 16S ribosomal RNA, complete sequence	Cronobacter turic...	2553	2553	100%	0.0	96.62%	1543	NR_102802.1
Shimwellia blattae DSM 4481 = NBRC 105725 16S ribosomal RNA, partial sequence	Shimwellia blatta...	2553	2553	100%	0.0	96.62%	1544	NR_074908.1
Lelliottia aquatilis strain 6331-17 16S ribosomal RNA, partial sequence	Lelliottia aquatilis	2551	2551	100%	0.0	96.62%	1536	NR_179968.1
Phytobacter palmae strain S29 16S ribosomal RNA, partial sequence	Phytobacter pal...	2547	2547	97%	0.0	97.58%	1487	NR_179034.1
Citrobacter koseri strain CDC-8132-86 16S ribosomal RNA, partial sequence	Citrobacter koseri	2547	2547	96%	0.0	97.89%	1469	NR_104890.1
Salmonella enterica subsp. houtenae strain DSM 9221 16S ribosomal RNA, partial sequence	Salmonella enter...	2545	2545	93%	0.0	98.61%	1437	NR_044371.1
Citrobacter braakii strain 167 16S ribosomal RNA, partial sequence	Citrobacter braakii	2545	2545	99%	0.0	96.73%	1530	NR_028687.1
Klebsiella pneumoniae strain DSM 30104 16S ribosomal RNA, partial sequence	Klebsiella pneum...	2543	2543	99%	0.0	96.79%	1524	NR_117682.1

Feedback

Huaxiibacter chinensis strain 155047 16S ribosomal RNA, partial sequence	Huaxiibacter chin...	2540	2540	100%	0.0	96.49%	1536	NR_184601.1
Cedecea lapagei strain DSM 4587 16S ribosomal RNA, partial sequence	Cedecea lapagei	2540	2540	100%	0.0	96.61%	1532	NR_126317.1
Enterobacter cloacae strain NBRC 13535 16S ribosomal RNA, partial sequence	Enterobacter clo...	2538	2538	95%	0.0	97.89%	1465	NR_113615.1
Kosakonia radicincitans DSM 16656 16S ribosomal RNA, partial sequence	Kosakonia radici...	2532	2532	100%	0.0	96.54%	1530	NR_117704.1
Phytobacter massiliensis JC163 16S ribosomal RNA, partial sequence	Phytobacter mas...	2531	2531	97%	0.0	97.13%	1499	NR_125600.1
Klebsiella aerogenes KCTC 2190 16S ribosomal RNA, complete sequence	Klebsiella aeroge...	2529	2529	100%	0.0	96.36%	1540	NR_102493.2
Erwinia tasmaniensis Et1/99 16S ribosomal RNA, partial sequence	Erwinia tasmanie...	2529	2529	100%	0.0	96.37%	1539	NR_074869.1
Kosakonia sacchari strain SP1 16S ribosomal RNA, partial sequence	Kosakonia sacchari	2529	2529	98%	0.0	97.08%	1500	NR_118333.1
Citrobacter koseri strain CIP 82.87 16S ribosomal RNA, partial sequence	Citrobacter koseri	2523	2523	95%	0.0	97.81%	1459	NR_118588.1
Enterobacter soli ATCC BAA-2102 strain LF7 16S ribosomal RNA, partial sequence	Enterobacter soli...	2523	2523	100%	0.0	96.23%	1535	NR_117547.1
Franconibacter dagui strain DL503 16S ribosomal RNA, partial sequence	Franconibacter d...	2521	2521	98%	0.0	96.95%	1503	NR_159235.1
Cedecea lapagei strain DSM 4587 16S ribosomal RNA, partial sequence	Cedecea lapagei	2521	2521	99%	0.0	96.41%	1531	NR_126319.1
Pantoea phytostimulans strain C3 16S ribosomal RNA, complete sequence	Pantoea phytosti...	2519	2519	100%	0.0	96.24%	1542	NR_199926.1
Cronobacter sakazakii strain ATCC 29544 16S ribosomal RNA, partial sequence	Cronobacter sak...	2519	2519	97%	0.0	97.13%	1493	NR_044076.1
Cedecea lapagei strain DSM 4587 16S ribosomal RNA, partial sequence	Cedecea lapagei	2518	2518	100%	0.0	96.35%	1531	NR_126318.1
Cronobacter muytjensii strain 05CHPL67 16S ribosomal RNA, partial sequence	Cronobacter muy...	2518	2518	98%	0.0	96.82%	1507	NR_117437.1
Escherichia albertii strain Albert 19982 16S ribosomal RNA, partial sequence	Escherichia alberti	2516	2516	97%	0.0	97.06%	1494	NR_025569.1
Escherichia marmotae strain HT073016 16S ribosomal RNA, partial sequence	Escherichia mar...	2512	2512	98%	0.0	96.81%	1504	NR_136472.1
Tatumella punctata strain SHS 2003 16S ribosomal RNA, partial sequence	Tatumella punctata	2510	2510	99%	0.0	96.34%	1527	NR_104937.1
Citrobacter freundii ATCC 8090 = MTCC 1658 = NBRC 12681 16S ribosomal RNA, partial sequence	Citrobacter freun...	2510	2510	98%	0.0	96.87%	1505	NR_028894.1
Citrobacter gillenii strain CDC 4693-86 16S ribosomal RNA, partial sequence	Citrobacter gillenii	2510	2510	99%	0.0	96.40%	1527	NR_041697.1
Shigella flexneri strain ATCC 29903 16S ribosomal RNA, partial sequence	Shigella flexneri	2510	2510	97%	0.0	97.11%	1488	NR_026331.1
Siccibacter turicensis LMG 23730 16S ribosomal RNA, partial sequence	Siccibacter turice...	2507	2507	97%	0.0	96.92%	1494	NR_104924.1
Klebsiella planticola ATCC 33531 strain JCM 7251 16S ribosomal RNA, partial sequence	Klebsiella planticola	2507	2507	94%	0.0	97.80%	1451	NR_112011.1
Pantoea stewartii subsp. indologenes strain CIP 104006 16S ribosomal RNA, partial sequence	Pantoea stewartii...	2505	2505	99%	0.0	96.33%	1524	NR_104928.1
Citrobacter pasteurii strain CIP 55.13 16S ribosomal RNA, partial sequence	Citrobacter paste...	2503	2503	97%	0.0	96.92%	1492	NR_178769.1
Shimwellia blattae DSM 4481 = NBRC 105725 strain CIP 104942 16S ribosomal RNA, partial sequence	Shimwellia blatta...	2503	2503	99%	0.0	96.33%	1525	NR_118256.1
Salmonella enterica subsp. arizonae strain DSM 9386 16S ribosomal RNA, partial sequence	Salmonella enter...	2501	2501	93%	0.0	98.05%	1438	NR_116125.1
Erwinia psidii strain LMG 7039 16S ribosomal RNA, partial sequence	Erwinia psidii	2499	2499	97%	0.0	96.85%	1494	NR_104569.1
Pectobacterium polaris strain NIBIO1006 16S ribosomal RNA, complete sequence	Pectobacterium ...	2497	2497	100%	0.0	95.98%	1542	NR_159084.1
Mixta intestinalis strain 29Y89B 16S ribosomal RNA, partial sequence	Mixta intestinalis	2497	2497	97%	0.0	96.79%	1502	NR_137380.1
Klebsiella pneumoniae strain DSM 30104 16S ribosomal RNA, partial sequence	Klebsiella pneum...	2497	2497	97%	0.0	96.79%	1492	NR_114715.1
Leclercia adecarboxylata strain NBRC 102595 16S ribosomal RNA, partial sequence	Leclercia adecar...	2494	2494	95%	0.0	97.35%	1465	NR_114154.1
Klebsiella oxytoca strain ATCC 13182 16S ribosomal RNA, partial sequence	Klebsiella oxytoca	2494	2494	98%	0.0	96.61%	1502	NR_118853.1
Pectobacterium polaris strain NIBIO1006 16S ribosomal RNA, complete sequence	Pectobacterium ...	2492	2492	100%	0.0	95.91%	1542	NR_159083.1
Klebsiella quasivariicola strain KPN1705 16S ribosomal RNA, partial sequence	Klebsiella quasiv...	2492	2492	98%	0.0	96.55%	1505	NR_181901.1
Erwinia mallotivora strain DSM 4565 16S ribosomal RNA, partial sequence	Erwinia mallotivora	2490	2490	98%	0.0	96.55%	1504	NR_041974.1
Pectobacterium polaris strain NIBIO1392 16S ribosomal RNA, partial sequence	Pectobacterium ...	2488	2488	100%	0.0	95.96%	1534	NR_159086.1
Mixta mediterraneensis strain Marseille-Q2057 16S ribosomal RNA, partial sequence	Mixta mediterran...	2486	2486	100%	0.0	95.85%	1538	NR_181487.1
Salmonella enterica subsp. indica strain DSM 14848 16S ribosomal RNA, partial sequence	Salmonella enter...	2486	2486	92%	0.0	98.38%	1414	NR_044370.1
Enterobacter cancerogenus strain LMG 2693 16S ribosomal RNA, partial sequence	Enterobacter can...	2486	2486	97%	0.0	96.53%	1495	NR_044977.1
Pectobacterium parmentieri strain RNS 08-42-1A 16S ribosomal RNA, complete sequence	Pectobacterium ...	2481	2481	100%	0.0	95.78%	1543	NR_153752.1
Pectobacterium quasiaquaticum strain A477-S1-J17 16S ribosomal RNA, complete sequence	Pectobacterium ...	2481	2481	100%	0.0	95.78%	1542	NR_181464.1
Buttiauxella agrestis ATCC 33320 strain DSM 4586 16S ribosomal RNA, partial sequence	Buttiauxella agre...	2481	2481	98%	0.0	96.43%	1504	NR_041968.1

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TYGS

TYGS vs 16S: TYGS is analyzing the entire genomes of organisms, whereas 16S BLAST is only looking at a single gene. TYGS also gives us the oppurtunity to build a phylogenetic tree based on the protein or DNA sequence.

Closest match + dDDH: Salmonella typhimurium JCM 1652

70% threshold: The 70% dDDH threshold means that if the dDDH value is equal to a greater than 70%, the identity of the species is the same as what is predicted. If the dDDH value is under 70%, then the unknown bacterium is likely to NOT be the same species as what was proposed.

Proteome tree option (low support): TYGS showing low branch support and offering a proteome-based tree implies that the evolutionary connections between the species on the tree are very close or there is not enough

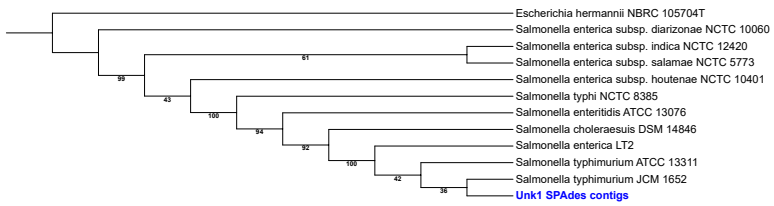
data to sufficiently draw a good proteome-based tree. The genomic data between species is either super close to where TYGS cannot tell the difference between species or the proteome difference is too large that TYGS tries to artificially fill in the gaps with predicted proteins.

iTOL tree

Tree adds value because: The tree is able to tell me the bootstrap values of multiple species whereas the single BLAST hit does not do that. From the bootstrap values, we can make predictions on how likely an evolutionary pathway was formed.

Support/trust: I am not too confident in the branch right next to the Unk1_SPAdes_contigs, which belongs to Salmonella typhimurium JCM 1652 because the bootstrap values for their relationship to their common ancestor are below 40%. That tells me that my unknown species and S. thyphimurium might not be that closely related as I thought. On the other hand, I did see a branch that had a bootstrap value of 100% between Salmonella enterica LT2 and the common ancestor for Salmonella typhimurium. Here the evolutionary relationship follows the same path every single time.

iTOL tree screenshot



Workflow note (for Git)

```
1. Filtered contigs shorter than 500bp; Tool: Filter FASTA (Galaxy Ver. 2.3) - Filtered on
sequence length minimum of 500bp

2. CheckM2; see genome completedness and contamination percentages; Tool: CheckM2 (Galaxy
Ver. 1.1.0+galaxy0), selected output quality prediction for both models for each genome in
Model Options

3. Prokka for genome annotation; Tool: Prokka (Galaxay Ver. 1.14.6+galaxy1), forced GenBank
compliance (Yes)

4. NCBI BLAST for species identification
Tool: NCBI BLAST - BLASTn (web), rRNA/ITS database

5. Build phylogenetic tree; Tool: Type Strain Genome Server (TSGS) - tygs.dsmz.de (web)
Parameters: Whole-Proteome tree

6. Visualize phylogenetic tree; Tool: interactive Tree of Life (iTOL) - iTOL v7 (web),
display bootstraps as text, rotated children, bold "Unk1_SPAdes contigs"

7. Avg. Nucleotide ID (ANI); Tool: EZ BioCloud (Web), from NCBI genome database, downloaded
the FASTA file for Salmonella Typhimurium (closest hit on NCBI 16S BLAST) and Salmonella
bongori (more distant relative). Compared filtered_contigs.fasta to the FASTA files from
Typhimurium and S. bongori
```